



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Dr Sukanya Ledalla
Student Name	Bhavana Kondakrindi
Roll Number	2520030082
Branch	CSE
Course Outcome	CO3 – Graph Traversal & Cycle Detection

Case Study Topic: CI/CD Workflow Dependency Management & Cycle Detection using DFS 3-Coloring Algorithm.

Work Done Summary:

In this case study, we implemented a Depth-First Search (DFS) algorithm using the 3-coloring scheme (White, Grey, Black) to analyze task dependencies in a continuous integration (CI) workflow pipeline. The system detects circular dependencies (back edges) that could cause infinite loops or deadlocks during a build execution.

Implementation Details:

1. Built a directed graph using an adjacency list (Tree Map) to manage jobs and their strict structural dependencies.
2. Implemented vertex state tracking via 3-coloring rules: WHITE (unvisited), GREY (currently visiting in the recursion stack), and BLACK (fully processed).
3. Developed an alphabetical-order neighbor traversal strategy to enforce a deterministic DFS path trace.
4. Programmed back-edge evaluation logic to trigger a cycle flag immediately upon encountering a GREY vertex target.
5. Calculated the overall topological runtime bounds ($O(V + E)$) governing active workflow validation.

Result: Screenshots/output



GitHub Link:

<https://github.com/bhavanakondakrindi/DSA-2-PROJECT/blob/main/CO3Assignment.java>

Student Signature	Faculty Signature
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